THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 6: **Materials Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student heats the ends of four rods P, Q, R, S, each with a wax-fixed pin at the same distance. The pins fall at 10 s, 25 s, 40 s and 60 s. The best conductor is: 1

(a) Q **(b)** S **(c)** P **(d)** R

2. A boy tests four materials A, B, C, D with a torch. Light passes fully through A, partly through B, not at all through C and D. Which can be used for a window to see out and still keep privacy? 1

(a) A **(b)** C **(c)** D **(d)** B

3. Why does a heavy ship made of steel float? 1

(a) water is sticky **(b)** steel is lighter than water
(c) ships have no weight **(d)** its shape displaces much water

4. In the experiment above, the worst conductor is: 1

(a) Q **(b)** P **(c)** R **(d)** S

5. If material C is an opaque metal and D is opaque wood, how would you distinguish them? 1

(a) C conducts heat and D does not **(b)** both float
(c) both transparent **(d)** both dissolve

6. Which of the following is a reason to use thermocol for packing glass items? 1

(a) it is hard **(b)** it is light and cushions shocks
(c) it is a metal **(d)** it is transparent

7. An iron cube and a wooden cube of the same size are placed in water. The iron sinks and the wood floats. What does this show? 1

(a) iron is heavier for the same size **(b)** wood is heavy
(c) wood is magnetic **(d)** iron is soluble

8. In an experiment, a mixture of sand, salt and iron filings is given. The first step of separation should be: 1

(a) adding water **(b)** filtering **(c)** using a magnet **(d)** heating

9. A bottle cap made of plastic is replaced by a metal cap. Which problem might arise? 1

(a) it becomes transparent **(b)** it floats
(c) it dissolves **(d)** it may rust or conduct heat

10. Which of these shows that solubility depends on the substance and not only on appearance? 1

(a) sugar and chalk powder both look white but behave differently in water
(b) both are white
(c) both are powders
(d) both are solid

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why an empty plastic bottle floats but a bottle filled with sand sinks. 2

12. Why do we choose rubber for the handle of an electrician's tools? 2

13. How can the same material produce different objects? 2

14. Why does a metal spoon in a cup of hot tea become hot but a plastic one hardly changes? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Substance X is a shiny solid that does not dissolve in water, sinks in water and becomes hot at the other end when one end is heated. Name its class of material and give two reasons. 3

16. Explain the importance of non-conducting handles in electrical tools. 3

17. Why are bells and utensils made of metals? 3

OR

Describe an experiment to show that air is a poor conductor of heat using materials of your choice. 3

18. Give three reasons why plastic has replaced many materials in daily life. 3

OR

A fabric absorbs 20 mL of water, another 5 mL. Which is better for a towel? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Design an investigation to rank wood, plastic, steel and copper for suitability as a cooking spoon handle. Name the variables and the safety steps. 5

OR

Describe an experiment to show that the same material in different forms (flat sheet and a boat shape) behaves differently in water. 5

20. A group of students wanted to prepare a separation plan for a mixture of iron filings, sand, salt and water-soluble blue dye. Write the full flow and explain the principle at each step. 5

OR

Explain how the properties of materials make a good umbrella and explain why each is chosen. 5

END OF QUESTION PAPER — Total: 40 Marks